

A DISTRIBUTION-FREE ANALYSIS OF VARIANCE
FOR THE TWO-WAY CLASSIFICATION

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1. Introduction and Summary

Amongst the earliest work on distribution-free methods of analysis of variance is that by Mood (1950) and Mood and Brown (1951) based on medians. For the two-way classification with b observations per cell, Mood (1950) discussed asymptotic distribution-free tests for interaction (the test for interaction is not completely distribution-free), for row-effects (assuming interaction to be zero) and for column-effects (assuming interaction to be zero). Mood and Brown proved that the different test statistics, considered one at a time, are under the null hypothesis of no effects asymptotically distributed as χ^2 for $b \rightarrow \infty$.

Mood and Brown's tests cannot be considered as very efficient since Andrews (1954) proved that the asymptotic relative efficiency (a.r.e.) of the Mood-Brown median test for the one-way classification compared with the F-test, for alternatives of shift in location, is $2/\pi$ in the case of normal distributed values and $1/3$ in the case of homogeneous distributed values.

Lehmann (1963a) presented a method to estimate the effects of the levels of any component in an analysis of variance, similar to the parametric case. In a subsequent paper (1963b) he presented a very general distribution-free method of analysis (still based on medians) for a one-, two- and multi-way classification. For the one-way classification there need not necessarily be an equal number of observations in each cell. The a.r.e. for alternatives of shift in location of Lehmann's (1963b) tests compared with the F-test is $3/\pi$, with the further advantage that all components in a two- or multi-way classification can be tested asymptotically independent of the outcome of the tests for the other components.

Lemmer and Stoker (1961) (cf. also Lemmer (1964)) presented a nonparametric analysis for a one-way classification which is a straight-forward generalisation of the well known Kruskal-Wallis test in that it is based on a nearly arbitrary function of the ranks instead of on the ranks themselves.

In this paper the theoretical derivation of a distribution-free analysis of variance for only a two-way classification is given since the generalisation from the two-way to the multi-way classification is immediate. To be more precise: A two factor classification consisting of k columns, m rows and b observations per cell is considered. The test statistics for the rows,

columns and interaction are obtained by partitioning the test statistic defined by Lemmer and Stoker (1961) into three test statistics for the three components. These three test statistics, which are based on a nearly arbitrary function of the ranks of the observations, are shown to be distributed under H_0 and certain general conditions, asymptotically independently as χ^2 variates for $b \rightarrow \infty$.

In Reinach's appendix to this paper the results of an investigation into the applicability of the formulae (derived under the null hypothesis) under alternative hypotheses in a special case of the function of the ranks are given and discussed.

2. Notation and Definitions

A two factor classification, consisting of k columns, m rows and b observations per cell, is considered. It is assumed that the b observations in the cell (i,j) (the cell formed by the i -th row and j -th column) are independently drawn from a population with distribution function $F_{ij}(x)$. The null hypothesis H_0 states that all km distribution functions $F_{ij}(x)$ are identical to, say, $F(x)$.

Consider N observations x_{ijh} , $i=1,2,\dots,m$, $j=1,2,\dots,k$, $h=1,2,\dots,b$ classified into m rows and k columns with b observations per cell ($N=mkb$).

All N observations are ranked together and ranks allocated, with r_{ijh} the rank of x_{ijh} . Let $Z=(z_1, z_2, \dots, z_N)$ denote the ranked observations where $z_1 \leq z_2 \leq \dots \leq z_N$. The theory of this paper is derived under the hypothesis H_{00} that all $N!$ permutations of the z_p 's ($p=1,2,\dots,N$) have equal probability. (Note that H_0 implies H_{00}).

Suppose the function $\Psi_N(\delta)$ is, for all N , continuous and monotonously increasing on the interval $0 \leq \delta \leq 1$ and that a continuous and strictly increasing function $\Psi(\delta)$ exists on the interval $0 < \delta < 1$ such that for all δ in $0 < \delta < 1$, $\lim_{N \rightarrow \infty} \Psi_N(\delta) = \Psi(\delta)$ uniformly

for $\epsilon > 0$, on $\epsilon \leq \delta \leq 1 - \epsilon$.

The observations x_{ijh} are now replaced by the numbers $\Psi_N[r_{ijh}/(N+1)]$ on which the test statistics are based. In the case of tied values among the observations the mean value of Ψ_N for each tie is allocated to the members of the tie, where the mean value is obtained by averaging in each tie the values of Ψ_N which the members of this tie would have had if they had been unequal (but otherwise in the same position with respect to all z_p not belonging to this tie).

Suppose λ ties of lengths $g_1, g_2, \dots, g_\lambda$ occur in this order among the ranked observations, where an observation different from all other observations is considered as a tie of length one.

Define $G_\alpha = g_1 + g_2 + \dots + g_\alpha$ with $\alpha \in \{1, 2, \dots, \lambda\}$
and $G_0 = 0$, and let

$$a_p = g_\alpha^{-1} \sum_{\mu=1}^{\alpha} \psi_N[(G_{\alpha-1} + \mu)/(N+1)] \quad \text{if } G_{\alpha-1} + 1 \leq p \leq G_\alpha,$$

with $a_{ijh} = a_p$ if $x_{ijh} = z_p$.

If all ties are of length one,

$$a_{ijh} = \psi_N[r_{ijh}/(N+1)] \quad \text{for all } i, j \text{ and } h,$$

and may be written as

$$a_p = \psi_N\left(\frac{p}{N+1}\right) = \psi_{Np}, \text{ say.}$$

Define*

$$a_{ij.} = b^{-1} \sum_h a_{ijh},$$

$$a_{i..} = k^{-1} \sum_j a_{ij.}, \quad a_{.j.} = m^{-1} \sum_i a_{ij.}$$

$$a_{...} = (mkb)^{-1} \sum_{ijh} a_{ijh} = N^{-1} \sum_p a_p = N^{-1} \sum_p \psi_{Np} \equiv \bar{\psi}_N,$$

$$\sigma_{Na}^2 = N^{-1} \sum_{ijh} (a_{ijh} - a_{...})^2 = N^{-1} \sum_p (a_p - a_{...})^2.$$

It is assumed that $\sigma_{Na}^2 > 0$, which will be the case for finite N
when at least two observations differ.

Let further

$$s_{ij} = C_N(a_{ij.} - a_{...})$$

$$t_{ij} = C_N(a_{ij.} - a_{i..} - a_{.j.} + a_{...})$$

$$t_{i.} = k^{-\frac{1}{2}} C_N(a_{i..} - a_{...})$$

$$t_{.j} = m^{-\frac{1}{2}} C_N(a_{.j.} - a_{...})$$

$$\text{where } C_N = (N-1)^{\frac{1}{2}} b^{-\frac{1}{2}} N^{-\frac{1}{2}} \sigma_{Na}^{-1}.$$

The following test statistics are now defined:

$$\text{For cell-effects : } T_T = \sum_{ij} s_{ij}^2$$

$$\text{For row-effects : } T_R = \sum_i t_{i.}^2$$

$$\text{For column-effects: } T_C = \sum_j t_{.j}^2$$

$$\text{For interaction : } T_I = \sum_{ij} t_{ij}^2.$$

3. Limiting Distributions of the Test Statistics

Theorem

If

$$(1) \quad \lim_{N \rightarrow \infty} \bar{\psi}_N \text{ exists,}$$

* Throughout this paper it is assumed that the summation indices
 $h, h' = 1, 2, \dots, b$; $i, i' = 1, 2, \dots, m$; $j, j' = 1, 2, \dots, k$ and
 $p, p' = 1, 2, \dots, N$.

$$(2) \quad \max_p (\psi_{Np} - \bar{\psi}_N)^2 = o(N),$$

$$(3) \quad N^{-1} \sum_p (\psi_{Np} - \bar{\psi}_N)^4 = o(N),$$

and

(4) (Every discontinuity of $F(x)$) $\leq \theta < 1$,
then the test statistics T_R , T_C and T_I have under H_{00} asymptotically, for $N \rightarrow \infty$, ^{**} independent χ^2 distributions with respectively $(m-1)$, $(k-1)$ and $(m-1)(k-1)$ degrees of freedom.

Proof

From the definitions of the test statistics it follows that

$$T_T = T_R + T_C + T_I.$$

Let $\vec{t}$ denote the column vector $\{t_1, \dots, t_m\}$ and A the $m \times m$ matrix $A = \{\delta_{ii} - \frac{1}{m}\}$ with δ_{ii} the Kronecker delta. We then have

$$\begin{aligned} T_R &= \sum_i t_i^2 \\ &= \vec{t}' A \vec{t} \end{aligned}$$

due to the linear relation $\sum_i t_i = 0$. The rank of A being $(m-1)$, it follows that T_R is a non-negative quadratic form of rank $(m-1)$ in the $t_i, i=1, \dots, m$, and since $t_i = k^{-1} \sum_j s_{ij}$, also in the $s_{ij}, i=1, \dots, m; j=1, \dots, k$.

Similarly, T_C is a non-negative quadratic form of rank $(k-1)$ and T_I a non-negative quadratic form of rank $(mk-1)$ in the $s_{ij}, i=1, \dots, m; j=1, \dots, k$.

Further, it can easily be shown, by making use of the relations $\sum_{ij} t_{ij} = \sum_j t_{ij} = 0$, that T_I is a non-negative quadratic form in the t_{ij} (and thus also in the s_{ij}), $i=1, \dots, m$ and $j=1, \dots, k$, with the $mk \times mk$ matrix of the form

$$\begin{aligned} E \equiv \{e_{ij, i'j'}\} &= \{\delta_{ii'} \delta_{jj'} - \frac{1}{m} \delta_{jj'} - \frac{1}{k} \delta_{ii'} + \frac{1}{km}\} \\ &= \{(\delta_{ii'} - \frac{1}{m})(\delta_{jj'} - \frac{1}{k})\}, \end{aligned}$$

which is readily seen to be of rank $(m-1)(k-1)$.

By making use of the result

$$\liminf_{N \rightarrow \infty} \sigma_{Na}^2 \geq \epsilon > 0, \quad \text{which holds with}$$

probability 1 under conditions (1) and (4) (cf. Stoker (1955) p. 34), it follows from an obvious extension of the derivation given in Lemmer and Stoker (1961) (where only the case of a continuous c.d.f. $F(x)$ was considered) that under H_{00} and conditions

(1) - (4) T_T has asymptotically, for $N \rightarrow \infty$, a χ^2 -distribution with $mk-1$ degrees of freedom (cf. Lemmer (1964)).

The theorem now follows from Cochran's theorem (cf. Scheffé

^{**}) m and k are finite numbers.

(1959) p. 420) and the remark given in Cramér (1946) p. 118. Note that the $mk \times mk$ matrix of the non-negative quadratic form T_T , viz. $\{\delta_{rr'} - \frac{1}{mk}\}$, $r, r' = 1, \dots, mk$, which is the same as the variance-covariance matrix of the s_{ij} , $i = 1, \dots, m$ and $j = 1, \dots, k$ (cf. Lemmer and Stoker (1961) p. 241), has $mk-1$ characteristic numbers equal to 1 and one characteristic number equal to zero (cf. Cramér (1946) p. 419); hence it follows from Cramér (1946) p. 313-314 that the above-mentioned remark is applicable.

4. Remarks

1) The above method is also applicable to the single degree of freedom approach, that is, T_T can be written as the sum of $mk-1$ test statistics corresponding to $mk-1$ orthogonal components in which the mk treatment combinations can be subdivided. Under H_{∞} and conditions (1) - (4) these $mk-1$ test statistics have asymptotically, for $N \rightarrow \infty$, independent χ^2 -distributions with one degree of freedom each.

2) As shown in Lemmer and Stoker (1961), test statistics of the Krushal-Wallis, Terry and Van der Waerden type can be obtained for the two-way classification by considering special cases of the function ψ_N .

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The test statistics T_R , T_C and T_I for the different components are not distributed asymptotically independent of each other under an alternative hypothesis, H_A , to the null hypothesis, H_0 . In a similar way as in Reinach (1965) for the randomized block design, it was shown by means of sampling studies, in the special case of test statistics of the Kruskal-Wallis type ($\psi_N(\delta) \equiv \delta$), that the tests for the different components become in general conservative under H_A (cf. Reinach (1966)).

The results obtained in the case of a 2×2 design with 6 replicates are given in Tables 1 and 2 for normal and exponential and homogeneous distributed values respectively, showing that increasing column-effects generally tend to decrease the values of the test statistics for the other components. This phenomenon can be explained in exactly the same way as that given in Reinach (1965) for the randomized block design (cf. Reinach (1966)).

Note that in the case of test statistics of the Kruskal-Wallis type the test statistic for, say, row-effects reduces to the well-known rank sum test if $m=2$ and to the Kruskal-Wallis test if

TABLE 1

Changes in the values of T_R and T_I with increasing column-effects in the case of random normal deviates for a 2×2 design with 6 replications. (The corresponding values of the F-statistics are also given).

T_R	T_C	T_I	F_R	F_C	F_I
0.053	2.613	1.080	0.030	1.745	2.466
0.030	6.163*	1.613	0.030	6.007*	2.466
0.120	8.670**	1.203	0.030	12.824**	2.466
0.083	10.830**	1.333	0.030	22.196**	2.466
0.053	15.413**	0.653	0.030	48.602**	2.466
0.030	16.803**	0.480	0.030	95.980**	2.466
3.413	1.920	1.080	4.362*	1.745	2.466
3.203	4.563*	1.613	4.362*	6.007**	2.466
2.253	7.363**	1.207	4.362*	12.824**	2.466
2.083	10.083**	1.333	4.362*	22.196**	2.466
1.470	14.963**	0.653	4.362*	48.602**	2.466
1.203	16.803**	0.480	4.362*	95.980**	2.466
14.083**	0.653	0.403	30.017**	1.745	2.466
12.209**	2.341	0.750	30.017**	6.007*	2.466
10.830**	4.083*	0.480	30.017**	12.824**	2.466
8.670**	6.453*	0.563	30.017**	22.196**	2.466
5.880*	11.603**	0.163	30.017**	48.602**	2.466
4.083*	15.413**	0.083	30.017**	95.980**	2.466

* Significant at the 5% level.

** Significant at the 1% level.

$m > 3$ with kb observations in each sample (assuming the effects of the other components to be zero).

Consequently, in view of the above phenomenon the tables of the exact probabilities of the rank sum test and the Kruskal-Wallis test can safely be used and the same applies to the small sample approximations of these test statistics.

The tests for the main components have undesirable consistency properties if the interaction effects are not zero. Reinach (1966) p. 264-265 illustrates by means of a few examples the possibility that interaction effects can lead to a significant outcome of a test for a main component although the effects of the levels of the main component have been taken to be zero.

An empirical investigation shows that it is not advisable to attach any value to the outcome of the test for a main component (say columns) if the ratio $T_I/T_C > \frac{1}{2}$ and to be careful if it is in the vicinity of a $\frac{1}{2}$ provided that the value of T_I is not significant. If the value of T_I is significant or if the ratio T_I/T_C is $> \frac{1}{2}$ and the test for the main component gives a significant

TABEL 2

Changes in the values of T_R and T_I with increasing column-effects for 2×2 designs with 6 replications.

Exponential distributed variates			Homogeneous distributed variates		
T_R	T_C	T_I	T_R	T_C	T_I
0.270	3.000	9.360**	0.120	0.480	15.413**
0.270	6.750**	7.680**	0.563	0.213	15.870**
0.213	9.720**	6.453*	0.163	2.430	14.520**
0.270	11.603**	5.333*	0.163	5.333*	11.603**
0.213	14.963**	3.630	0.013	11.213**	7.680**
0.120	16.803**	3.203	0.013	14.520**	5.333*
3.203	3.203	6.453*	13.653**	0.013	5.880*
3.000	5.070*	5.070*	13.653**	0.003	6.163*
2.253	7.363**	4.083*	12.813**	0.120	5.333*
1.470	10.453**	3.203	12.000**	0.750	3.630
1.080	14.520**	1.920	9.363**	3.630	1.613
0.750	16.803**	1.613	7.680**	7.053**	0.653
7.680**	1.763	4.563*	17.280**	0.013	3.853*
7.363**	2.803	3.413	17.280**	0.003	4.083*
6.453*	4.320*	2.613	17.280**	0.053	3.413
5.070*	6.750**	1.920	17.280**	0.403	2.083
3.630	11.213**	0.963	16.803**	1.763	0.653
2.083	15.413**	0.750	14.963**	4.083*	0.120

* Significant at the 5% level.

** Significant at the 1% level.

outcome, the treatment combinations are to be considered as a one-way classification and can be compared with each other by means of multiple comparison procedures.

Remark

A similar result holds for the randomized block design (Reinach (1965)) in which case it does not seem advisable to attach any value to the outcome of the test for a main component if the ratio $T_{\text{interaction}} / T_{\text{main component}} > 1$.

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